

SATAG: SUPERPOSITION-AWARE TEMPORAL AUDIO GENERATION

JinHyeong Kim^{1,†}, DongHwan Jang¹, Ji-Hwan Kim¹

¹Sogang University, Seoul, Republic of Korea

ABSTRACT

[TODO: Write the abstract; the Word source contains only template instructions.]

Index Terms— [TODO: Add keywords; the Word source contains example keywords only.]

1. INTRODUCTION

Text-to-Audio (TTA) generation has rapidly advanced toward synthesizing diverse environmental sounds, sound effects, and complex acoustic scenes from natural-language descriptions. Recent studies have extended this capability to temporally controllable TTA, where users can explicitly specify when individual sound events occur and how long they persist. For example, DegDiT represents sound events and their temporal relations using a graph structure to control event timing, and PicoAudio2 conditions generation on event-level timestamps to specify the onset and offset of multiple sound events. These approaches provide fine-grained temporal control, but generation becomes less reliable when multiple events overlap in time.

We observe several characteristic failure behaviors under temporal overlap, including Event Omission, Event Fusion, and Temporal Collapse. Fig. 1 illustrates these behaviors using the same event pair under different timestamp configurations. When the two events occupy separate temporal regions, each event is typically generated with a clear identity. As temporal overlap increases, both event identity and the specified temporal structure become more difficult to preserve.

Under the linear acoustics assumption, overlapping sound sources form an acoustic mixture through the linear superposition of their pressure signals, which corresponds to sample-wise addition in the digital waveform domain. Overlapping-event generation therefore requires multiple event-specific signals to coexist within the same waveform segment. Existing temporally controllable TTA models process multiple event conditions within a single generative trajectory and directly predict a joint audio representation. We find that this joint generation process becomes particularly vulnerable when multiple event-specific structures must be represented within the same temporal interval.

Motivated by this observation, we introduce SATAG (Superposition-Aware Temporal Audio Generation), a training-free inference framework for temporally controllable TTA. SATAG detects temporal overlap from the onset and offset conditions of the input events and applies waveform-domain acoustic superposition during inference. SATAG also supports event-level mixing control, allowing the relative contribution of each event to the final waveform mixture to be explicitly specified. It requires no additional training, parameter updates, or architectural modifications to the underlying TTA model.

Our contributions are as follows: (1) we identify acoustic superposition as a key factor underlying generation failures in temporally

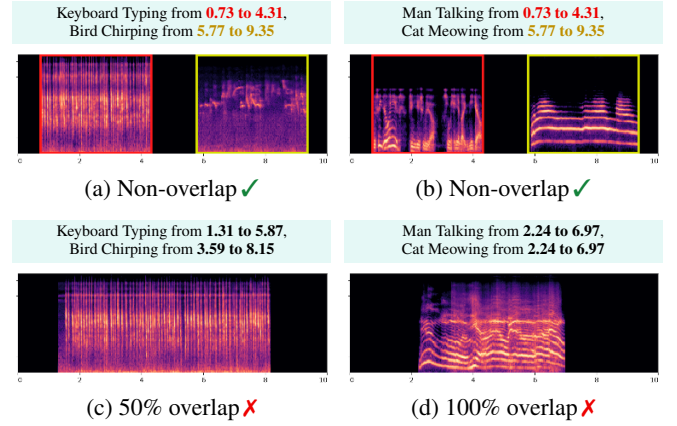


Fig. 1. Examples of generated audio under different temporal overlap ratios.

controllable TTA under temporal overlap; (2) we propose SATAG, a training-free superposition-aware inference method for overlapping-event generation with event-level mixing control; and (3) we demonstrate improvements in event occurrence, temporal alignment, and perceptual quality across different temporal-overlap conditions.

2. SUPERPOSITION-AWARE TEMPORAL AUDIO GENERATION

2.1. Background

Sound superposition.

In acoustics, multiple sound waves propagating through the same space are combined according to the principle of superposition. Under the linear acoustics assumption, the total acoustic pressure at a given observation point is expressed as the sum of the acoustic pressures generated by the individual sources. Let $p_i(t)$ denote the acoustic pressure generated by the i -th source. The resulting acoustic pressure $p_{\text{mix}}(t)$ is given by

$$p_{\text{mix}}(t) = \sum_{i=1}^N p_i(t). \quad (1)$$

For two sound sources, this relation becomes

$$p_{\text{mix}}(t) = p_1(t) + p_2(t). \quad (2)$$

In digital audio, acoustic pressure is represented as a discrete waveform after sampling. Let $x_i[n]$ denote the discrete waveform of the i -th source, where n denotes the sample index. Assuming that all source signals share the same sampling rate and temporal reference, waveform-domain mixing is expressed as

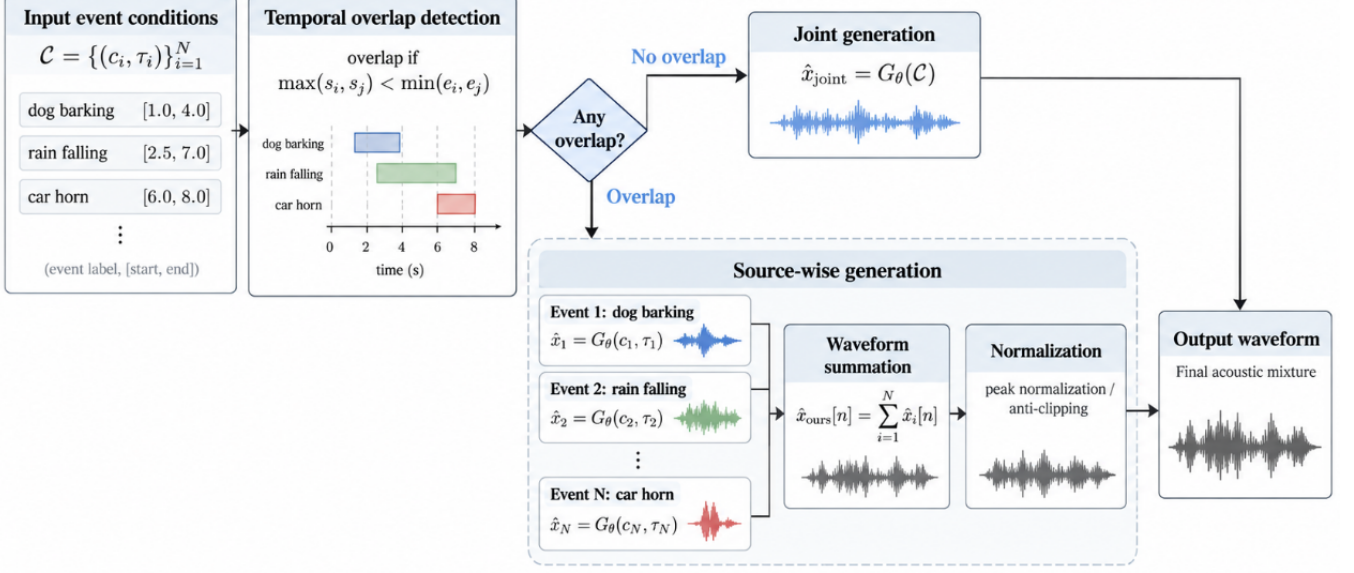


Fig. 2. Overall architecture of SATAG.

$$x_{\text{mix}}[n] = \sum_{i=1}^N x_i[n]. \quad (3)$$

Accordingly, when multiple sound events overlap in time, the waveform samples within the overlapping interval are formed by the sample-wise summation of the corresponding source signals. This additive property provides the physical basis for constructing an audio mixture from overlapping sound sources.

2.2. SATAG Framework

We propose SATAG (Superposition-Aware Temporal Audio Generation), a training-free inference framework that incorporates acoustic superposition into temporally controllable TTA. Given a set of event conditions $\mathcal{C} = \{(c_i, \tau_i)\}_{i=1}^N$, where c_i denotes the semantic condition of the i -th event and $\tau_i = (s_i, e_i)$ denotes its onset and offset, SATAG first identifies temporal overlap between event intervals. Events i and j are considered overlapping when

$$\max(s_i, s_j) < \min(e_i, e_j). \quad (4)$$

For inputs without temporal overlap, SATAG retains the original generation procedure. When temporal overlap is detected, event-specific waveforms are obtained from the pretrained temporally controllable TTA model,

$$\hat{x}_i = G_{\theta}(c_i, \tau_i), \quad (5)$$

where G_{θ} denotes the generation model with fixed parameters. These event conditions can be processed in parallel along the batch dimension. SATAG then constructs the overlapping acoustic mixture in the waveform domain according to the superposition principle introduced in Sec. 2.1.

When no event-level mixing condition is specified, the resulting waveform is formulated as

$$x_{\text{mix}}[n] = \sum_{i=1}^N \hat{x}_i[n]. \quad (6)$$

To maintain the output waveform within the valid amplitude range, a global peak-normalization operator is defined as

$$P(x) = \frac{x}{\max_n |x[n]| + \epsilon}, \quad (7)$$

and the final output is obtained by

$$\hat{x}_{\text{SATAG}} = P(x_{\text{mix}}). \quad (8)$$

SATAG further supports event-level mixing control by introducing a relative mixing condition $r_i > 0$ for each event. Since the generated event waveforms may have different amplitude scales, each waveform is first mapped to a common peak-amplitude reference,

$$\tilde{x}_i = P(\hat{x}_i). \quad (9)$$

The specified mixing condition is then incorporated into the superposition process as

$$x_{\text{mix}}[n] = \sum_{i=1}^N r_i \tilde{x}_i[n], \quad (10)$$

followed by global peak normalization,

$$\hat{x}_{\text{SATAG}} = P(x_{\text{mix}}). \quad (11)$$

The common scaling applied to the final mixture preserves the relative relationship specified by r_i . In this way, SATAG incorporates acoustic superposition into overlapping event generation while extending timestamp-based control with event-level mixing control, without additional training or modification of the underlying TTA model.

3. EXPERIMENTS

3.1. Dataset

Temporally controllable TTA requires audio–text pairs with event-level temporal annotations. However, existing temporally annotated audio datasets often provide only coarse class labels rather than fine-grained event descriptions. For example, datasets such as Audio-Condition and AudioSet-Strong provide temporally strong annotations based on a fixed set of sound event classes. As a result, an audio event with a more specific acoustic action, such as dog barking, may be annotated only with a categorical label such as dog, without explicitly describing the associated action. Such class-level annotations provide accurate temporal boundaries but limited event descriptions, restricting the amount of event-specific semantic information available for training temporally controllable TTA models. To address this limitation, we construct AudioCaps-T by augmenting the fine-grained event descriptions in AudioCaps with automatically estimated onset and offset timestamps. We first use a large language model to extract individual sound events from each caption, which are then associated with their corresponding temporal boundaries. We then apply SpotSound to the corresponding audio clip to estimate the onset and offset timestamps of each extracted sound event. This process transforms the original caption-level annotations into event-level temporal annotations, represented as $\mathcal{A} = \{(c_i, s_i, e_i)\}_{i=1}^N$, where c_i denotes the i -th sound event and s_i and e_i denote its estimated onset and offset, respectively.

[TODO: Add the dataset subjective evaluation result.]

3.2. Experiment Setup

We use DegDiT as the baseline model for all main experiments. DegDiT explicitly represents sound events and their temporal information in a graph-based structure, making it suitable for evaluating temporal control under overlapping-event conditions and for applying SATAG without modifying the underlying architecture.

We train DegDiT on AudioCaps-T for 100 epochs. The model is optimized with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 1e-8$, weight decay 0.01) using an initial learning rate of 3×10^{-4} , a batch size of 4 per GPU, effective 16 (4 GPU, grad accum 1), and linear warmup + linear decay to 0 (`transformers.get_scheduler("linear")`), warmup 400 steps.

For evaluation, we construct a controlled event set from AudioCaps-T to reduce failures caused by insufficient training exposure. We first retain events that occur at least 30 times in the dataset and further require at least 20 isolated occurrences, where the event appears without temporal overlap with other events. These criteria ensure that the selected events are sufficiently represented during training and have been observed independently. The filtering process results in 354 unique events, corresponding to 62,481 possible event pairs. From these pairs, we randomly sample 10,000 pairs and evaluate DegDiT and SATAG using identical event combinations and temporal conditions under 50% and 100% temporal overlap.

3.3. Evaluation Metrics

We evaluate temporal generation performance using Event Recall, Onset MAE, and Offset MAE with SpotSound. Following the evaluation report, Event Recall measures the mean number of conditioned events per sample whose predicted intervals are fully contained within their target intervals. Let $N_{\text{contained}}$ denote the total number of such events and N_{samples} denote the number of evaluated samples. Event Recall is defined as

Method	Event Recall $\uparrow$	Onset MAE $\downarrow$	Offset MAE $\downarrow$
50% temporal overlap			
DegDiT	0.5690	0.8138	1.1722
SATAG	0.8054	0.6980	1.0942
100% temporal overlap			
DegDiT	0.4407	0.8225	0.8822
SATAG	0.1513	0.5380	0.6754

Table 1. Temporal generation performance under 50% and 100% overlap. Event Recall is the mean contained-event count per sample (0–2); MAE is in seconds. Best values within each overlap condition are in bold.

$$\text{Event Recall} = \frac{N_{\text{contained}}}{N_{\text{samples}}}.$$

For the two-event samples used in this evaluation, Event Recall ranges from 0 to 2.

To evaluate temporal accuracy, we measure the absolute difference between the target timestamps and the detected timestamps. For each sample, Onset MAE and Offset MAE are computed over events with valid predicted intervals as

$$\text{Onset MAE} = \frac{1}{M} \sum_{i=1}^M |\hat{s}_i - s_i|,$$

$$\text{Offset MAE} = \frac{1}{M} \sum_{i=1}^M |\hat{e}_i - e_i|,$$

where s_i and e_i denote the target onset and offset timestamps, respectively, while $\hat{s}_i$ and $\hat{e}_i$ denote the corresponding detected timestamps. M denotes the number of events with valid predicted intervals in the sample. The reported MAE values are the means of these per-sample errors, excluding samples without valid predictions. Lower MAE indicates more accurate temporal alignment.

We additionally conduct a subjective evaluation to assess both perceptual quality and temporal controllability. MOS evaluates the overall naturalness and perceptual quality of the generated audio, whereas Temporal Alignment MOS evaluates how well the generated sound events follow the specified onset and offset conditions. We evaluate these two aspects separately to examine whether improved temporal controllability is achieved without degrading perceptual audio quality.

3.4. Results

Table 1 compares DegDiT and SATAG under 50% and 100% temporal-overlap conditions. At 50% overlap, SATAG improves Event Recall from 0.5690 to 0.8054, while reducing Onset MAE from 0.8138 s to 0.6980 s and Offset MAE from 1.1722 s to 1.0942 s. Under 100% overlap, Onset MAE decreases from 0.8225 s to 0.5380 s and Offset MAE decreases from 0.8822 s to 0.6754 s, while Event Recall decreases from 0.4407 to 0.1513. Thus, SATAG improves temporal accuracy for events with valid predictions under both overlap conditions, but its improvement in Event Recall is limited to 50% overlap.

[TODO: Qualitative figure] presents qualitative examples of generated audio under temporal overlap. DegDiT exhibits representative failure behaviors such as Event Omission, Event Fusion, and

Method	Inference Time ↓ (s)	RTF ↓	Peak Memory ↓
DegDiT	1.7247	0.1725	5.308
SATAG	2.936	0.2940	5.308

Table 2. Computational efficiency comparison of DegDiT and SATAG. Best values are in bold, including ties.

Temporal Collapse, whereas SATAG produces more distinguishable event structures while maintaining the requested temporal intervals. The corresponding mel-spectrograms further illustrate the improved preservation of individual events within the overlapping region.

Subjective evaluation results are reported in [TODO: Subjective evaluation table]. SATAG achieves a MOS of [TODO], compared with [TODO] for DegDiT, and improves Temporal Alignment MOS from [TODO] to [TODO]. The results indicate that the improvement in temporal controllability does not degrade the perceptual quality of the generated audio.

3.5. Computational Efficiency

Although SATAG processes overlapping events separately, the corresponding generations are executed in parallel along the batch dimension to reduce inference latency. As shown in Table 2, SATAG requires 2.936 s per sample with an RTF of 0.2940, compared with 1.7247 s and an RTF of 0.1725 for DegDiT, while requiring no additional training or parameter updates.

4. CONCLUSION

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5. REFERENCES

[TODO: Add verified references and in-text citations; the Word source contains only formatting instructions and fictitious example references.]